

TABLE V
COMPUTERIZATION AND INDUSTRY TASK INPUT 1980–1998:
OVERALL AND BY EDUCATION GROUP

DEPENDENT VARIABLE: $10 \times$ ANNUAL CHANGE IN QUANTILES OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

	1. Δ Nonroutine analytic	2. Δ Nonroutine interactive	3. Δ Routine cognitive	4. Δ Routine manual
A. Aggregate within-industry change				
Δ Computer use	12.95	15.97	-15.84	-14.32
1984–1997	(3.68)	(4.32)	(4.73)	(4.73)
Intercept	-0.33	1.27	0.38	0.54
	(0.77)	(0.90)	(0.99)	(0.99)
Weighted mean task Δ	2.20	4.39	-2.71	-2.25
B. Within industry: High school dropouts				
Δ Computer use	4.64	11.92	-2.64	-8.85
1984–1997	(6.07)	(8.73)	(7.95)	(6.76)
Intercept	-2.51	-4.39	0.02	1.11
	(1.26)	(1.82)	(1.66)	(1.41)
Weighted mean task Δ	-1.61	-2.07	-0.49	-0.62
C. Within industry: High school graduates				
Δ Computer use	0.04	13.49	-28.18	-25.50
1984–1997	(4.17)	(5.40)	(6.13)	(6.05)
Intercept	-1.49	1.07	1.55	0.48
	(0.87)	(1.13)	(1.28)	(1.26)
Weighted mean task Δ	-1.48	3.70	-3.95	-4.49
D. Within industry: Some college				
Δ Computer use	7.95	18.14	-15.68	-17.77
1984–1997	(5.03)	(5.54)	(5.27)	(5.61)
Intercept	-1.88	-0.58	0.35	1.39
	(1.05)	(1.15)	(1.10)	(1.17)
Weighted mean task Δ	-0.33	2.96	-2.71	-2.08
E. Within industry: College graduates				
Δ Computer use	1.61	5.57	-0.78	-4.46
1984–1997	(3.42)	(3.35)	(4.85)	(5.70)
Intercept	0.25	0.10	-0.96	-0.12
	(0.71)	(0.70)	(1.01)	(1.19)
Weighted mean task Δ	0.57	2.22	-1.48	-1.98
F. Decomposition into within and between education group components				
Explained task Δ	2.52	3.11	-3.09	-2.79
Within educ groups (%)	23.7	77.9	91.7	111.1
Between educ groups (%)	76.3	22.1	8.3	-11.1

n in panels A–E is 140, 139, 140, 140, and 139 consistent CIC industries. Standard errors are in parentheses. Each column of panels A–E presents a separate OLS regression of ten times the annual change in industry-level task input for the relevant education group (measured in centiles of the 1960 task distribution) during 1980–1998 on the annual percentage point change in industry computer use during 1984–1997 (weighted mean 0.198) and a constant. Estimates are weighted by mean industry share of total employment (in FTEs) in 1980 and 1988. Industries with no employment in the relevant educational category in either 1980 or 1998 are excluded. Data sources are CPS MORG 1980 and 1998 and DOT 77 job task measures. The “explained” component in Panel F is the within-industry change in the task measure predicted by computerization in regression models in Panel A. See Table I and Appendix 1 for definitions and examples of task variables.

bution—college graduates and high school dropouts—similar patterns prevail, but they are less precisely estimated. In all cases, the estimates are of the expected sign, but none is statistically significant. For college graduates, this is likely to reflect “topping out,” since this education group was already at the extreme of the distribution for all tasks. We are less certain why the relationships are weaker for high school dropouts, but one possibility is that this group has insufficient human capital to be effectively redeployed to alternative job tasks.

To assess whether these within-education group shifts are a quantitatively important component of the overall change in industry task content, panel F of Table V presents a decomposition of industry task changes into within and between education group components. This exercise shows that in every case, within-education group task upgrading explains a substantial share, 24 to 111 percent, of total task upgrading over these two decades. For example, the annual within-industry change in nonroutine interactive tasks over the period 1980–1998 is 4.4 centiles per decade, of which 3.1 centiles (71 percent) is accounted for by contemporaneous industry computerization. Within-education group task changes explain the bulk of these shifts: 78 percent of the explained component and 55 percent of the total. Subdividing the explained within-education group component further, 59 percent is due to changes in task assignment among high school graduates and those with some college, and the rest is equally accounted for by task shifts among college graduates and high school dropouts.

This exercise demonstrates that within-education group shifts in task content are the primary channel through which the structure of workplace tasks has shifted over the past two decades. Furthermore, a large portion of the within-education group changes are accounted for by cross-industry patterns of computer adoption. This suggests to us that task change is antecedent to educational upgrading, rather than merely a reflection of it.

V.B. Task Shifts within Occupations

The analyses above exploit shifts in occupational composition—the extensive margin—to quantify changes in task input. This approach is imperfect since it assumes that the tasks performed within occupations are static, which is unlikely to be accurate over long time intervals. Moreover, our task framework implies that this assumption should be violated in a specific

manner: occupations undergoing rapid computerization should differentially reduce labor input of routine cognitive and manual tasks and increase labor input of nonroutine cognitive tasks. To provide one example, the 1976 edition of the Department of Labor's *Occupation Outlook Handbook* described the job of Secretary as: "... Secretaries relieve their employers of routine duties so they can work on more important matters. Although most secretaries type, take shorthand, and deal with callers, the time spent on these duties varies in different types of organizations" [U. S. Department of Labor 1976, p. 94]. In 2000 the entry for Secretary reads: "As technology continues to expand in offices across the Nation, the role of the secretary has greatly evolved. Office automation and organizational restructuring have led secretaries to assume a wide range of new responsibilities once reserved for managerial and professional staff. Many secretaries now provide training and orientation to new staff, conduct research on the Internet, and learn to operate new office technologies" [U. S. Department of Labor 2000, p. 324].

To test whether this example captures a pervasive phenomenon, we match occupations from the 1977 and 1991 revisions of the DOT to estimate the following equation:

$$(16) \quad \Delta T_{mk\tau} = \alpha + \xi \Delta C_m + \varepsilon_{mk\tau}.$$

Here, $\Delta T_{mk\tau}$ is the change in occupational input of task k between 1977 and 1991 in three-digit COC occupation m , and ΔC_m is the change in occupational computer penetration measured by the CPS. To provide a clean test, our data set is constructed using only the subset of occupations appearing in the 1977 DOT, which was used to create our original occupation crosswalk. Accordingly, the variation used to estimate equation (16) stems exclusively from DOT examiners' reevaluation of the task content of individual occupations between 1977 and 1991.²³

Table VI presents three estimates for each task measure. The first column of each panel presents a bivariate regression of the within-occupation change in task content on occupational computerization and a constant. These estimates provide striking

23. The weighted fraction of employment reevaluated between 1978 and 1990 in our data is 73 percent. Occupations were chosen for reevaluation by DOT examiners partly on the expectation that their content had changed. Hence, this is not a random sample. We assume that occupations that were not revised between the 1977 and 1991 DOT experienced no task change. Provided that these occupations did not experience offsetting shifts, our approach will provide a lower bound on the extent of task change.

TABLE VI
COMPUTERIZATION AND CHANGES IN JOB TASK CONTENT WITHIN OCCUPATIONS 1977-1991
DEPENDENT VARIABLE: $10 \times$ ANNUAL WITHIN-OCCUPATION CHANGE IN QUANTILE OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1984 TASK DISTRIBUTION

	A. Δ Nonroutine analytic			B. Δ Nonroutine interactive			C. Δ Routine cognitive			D. Δ Routine manual		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Δ Computer use 1984-1997	2.94 (1.84)	3.57 (1.92)	4.02 (2.06)	5.70 (1.88)	5.86 (1.97)	7.08 (2.11)	-18.18 (3.29)	-16.56 (3.41)	-18.48 (3.65)	1.74 (2.89)	0.83 (3.01)	0.37 (3.23)
Δ College grad emp. 1984-1997		-4.79 (5.54)	-4.83 (5.54)		-4.47 (5.68)	-4.58 (5.67)		22.59 (9.86)	22.76 (9.85)		-16.07 (8.70)	-16.03 (8.71)
Δ HS grad emp. 1984-1997		2.83 (3.78)	3.09 (3.81)		-0.19 (3.88)	0.52 (3.90)		16.97 (6.73)	15.86 (6.77)		-10.42 (5.94)	-10.70 (5.99)
Δ Female emp. 1984-1997			-2.37 (3.94)			-6.47 (4.03)			10.14 (6.99)			2.47 (6.19)
Intercept	-0.92 (0.40)	-0.91 (0.41)	-0.95 (0.41)	-0.46 (0.41)	-0.42 (0.42)	-0.52 (0.42)	0.56 (0.71)	0.14 (0.72)	0.30 (0.73)	0.42 (0.63)	0.70 (0.64)	0.74 (0.64)
R^2	0.01	0.01	0.01	0.02	0.02	0.03	0.06	0.08	0.08	0.00	0.01	0.01
Weighted mean Δ		-0.39			0.58			-2.76			0.74	

n is 470 consistent three-digit Census Occupation Code (COC) occupations. Standard errors are in parentheses. Each column presents a separate OLS regression of ten times the annual change in the occupational task measure (measured in centiles of the 1984 distribution) between the 1977 and 1991 DOT revisions. Computer use, college graduate, high school graduate, and female employment shares are measured as ten times the annual change in the relevant measure from the 1984 and 1997 CPS. Weighted means are 0.183, 0.017, -0.015, and 0.017, respectively. Omitted education categories are some college and high school dropout. Estimates are weighted by the average occupational share of U. S. employment in 1984 and 1997. See Table I and Appendix 1 for definitions and examples of task variables.